

LEEDS BRADFORD, United Kingdom

WMO: 033463

Lat: 53.866N

Lon: 1.661W

Elev: 208

StdP: 98.86

Time Zone: 0.00 (EUW)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.2	-2.1	-5.2	2.5	-0.5	-4.2	2.7	-0.1	17.3	8.4	14.8	7.0	3.6	280	0.665

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	24.0	17.8	22.0	16.7	20.1	15.8	18.5	22.3	17.4	20.8	16.5	19.5	4.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.5	20.2	16.0	11.7	18.8	15.1	11.0	17.7	53.2	22.1	49.9	20.9	47.1	19.3	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.7	11.0	9.6	-5.3	26.8	1.6	1.8	-6.5	28.0	-7.5	29.1	-8.4	30.1	-9.6	31.3
			-5.4	21.2	1.7	1.9	-6.7	22.5	-7.7	23.6	-8.6	24.7	-9.9	26.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	3.7	4.0	5.4	7.6	10.5	13.2	15.4	15.1	12.8	9.7	6.1	4.1
	DBStd	5.04	2.88	3.00	2.92	2.85	2.91	2.57	2.49	2.33	2.27	2.73	2.86	3.24
	HDD10.0	963	194	168	144	82	29	3	0	0	3	38	121	182
	HDD18.3	3418	452	401	401	323	243	154	98	104	167	268	368	440
	CDD10.0	600	0	1	2	9	44	100	167	159	87	29	3	1
	CDD18.3	13	0	0	0	0	0	2	6	4	1	0	0	0
	CDH23.3	81	0	0	0	0	1	12	41	24	2	1	0	0
	CDH26.7	6	0	0	0	0	0	1	3	2	0	0	0	0
Wind	WSAvg	4.7	5.4	5.5	5.1	4.6	4.4	4.2	4.2	4.1	4.2	4.6	4.9	5.2
Precipitation	PrecAvg	877	90	69	72	63	59	67	55	69	74	76	88	92
	PrecMax	1112	180	168	140	131	124	196	140	161	210	171	169	195
	PrecMin	646	6	7	13	8	8	11	8	7	7	13	34	31
	PrecStd	121	41	39	33	34	31	43	31	32	47	41	42	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	13.1	14.2	19.0	22.8	25.2	26.8	26.2	22.9	18.2	14.1	12.2
		MCWB	10.4	10.2	10.1	12.9	16.6	18.5	18.5	17.8	17.8	15.2	12.7	10.8
	2%	DB	10.1	11.0	12.2	15.9	20.0	22.0	24.1	23.0	19.9	16.0	12.8	11.1
		MCWB	9.0	9.2	9.3	11.3	14.7	16.5	17.9	17.4	15.7	14.0	11.5	10.0
	5%	DB	9.1	9.8	11.0	13.9	17.2	19.9	22.1	20.8	18.0	14.9	11.2	10.0
		MCWB	8.4	8.5	8.6	10.2	12.9	15.2	16.9	16.6	14.8	13.2	10.0	9.2
	10%	DB	8.1	8.2	9.9	12.2	15.2	18.0	20.1	19.1	16.8	13.9	10.2	9.0
		MCWB	7.3	7.2	8.1	9.0	11.7	14.0	16.0	15.5	13.8	12.3	9.3	8.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	11.0	11.3	14.0	17.1	19.5	20.0	19.3	18.2	16.0	13.2	11.6
		MCDB	11.5	12.2	13.4	16.5	21.7	23.6	24.6	23.2	21.2	17.9	14.2	12.4
	2%	WB	9.4	9.6	10.1	11.9	14.9	17.1	18.5	18.0	16.5	14.4	11.9	10.5
		MCDB	10.0	10.6	11.6	15.0	18.7	20.6	22.7	21.2	19.0	15.7	12.7	11.2
	5%	WB	8.4	8.4	9.1	10.7	13.6	15.7	17.4	17.1	15.4	13.4	10.7	9.3
		MCDB	9.0	9.4	10.4	13.3	16.8	18.8	21.1	20.3	17.4	14.6	11.4	9.8
	10%	WB	7.3	7.3	8.1	9.6	12.3	14.6	16.4	16.2	14.4	12.4	9.5	8.1
		MCDB	7.8	8.2	9.6	11.8	15.0	17.4	19.6	18.7	16.1	13.6	10.2	8.8
Mean Daily Temperature Range	5% DB	MDBR	4.1	4.8	5.9	7.1	7.5	7.4	7.5	7.0	6.4	5.4	4.6	4.3
		MCDBR	5.4	5.9	7.4	9.4	10.5	10.7	10.8	9.4	8.3	6.3	5.3	5.4
		MCWBR	5.3	4.9	5.0	5.8	6.4	6.4	5.7	5.0	5.1	4.7	5.1	5.1
	5% WB	MCDBR	5.3	5.5	6.7	8.6	9.9	9.5	9.7	8.3	7.4	6.0	5.3	5.4
		MCWBR	5.4	5.0	5.2	5.9	6.4	6.3	5.8	5.0	5.3	5.0	5.3	5.3
Clear-Sky Solar Irradiance	taub	0.314	0.329	0.358	0.384	0.384	0.387	0.387	0.374	0.356	0.339	0.314	0.303	
	taud	2.331	2.388	2.314	2.260	2.296	2.313	2.314	2.367	2.426	2.448	2.397	2.311	
	Ebn at Noon	643	767	817	842	862	862	854	844	815	746	643	575	
	Edh at Noon	58	75	99	119	121	121	119	106	89	70	54	49	
All-Sky Solar Radiation	RadAvg	0.63	1.29	2.39	3.73	4.73	4.86	4.70	3.87	2.82	1.58	0.78	0.49	
	RadStd	0.07	0.15	0.28	0.42	0.30	0.45	0.46	0.25	0.25	0.14	0.08	0.05	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page